REVIEW 1 · PLAIN-LANGUAGE BRIEFING SHEET

What we are building, and why

Satellite-Based Early Warning for Glacial Lake Outburst Floods · Major Project, 7th Semester · Department of Artificial Intelligence & Machine Learning

In one sentence: we are building software that keeps an eye on dangerous mountain lakes using free satellite pictures, and warns an officer when a lake starts behaving strangely.

1. What is the issue?

High in the Himalaya, glaciers are melting. As a glacier retreats, the meltwater collects in a hollow and forms a lake. The wall holding that water back is not a proper dam — it is just a heap of loose rock, mud and buried ice that the glacier left behind. Nobody built it. Nobody checks it.

Sometimes that wall gives way. When it does, the whole lake empties in a few hours. Millions of tonnes of water, rock and mud rush down the valley and destroy everything in the path — villages, roads, bridges, power stations. This is called a **Glacial Lake Outburst Flood**, or GLOF.

Two things make this worse every year. Warming melts more ice, so these lakes are getting bigger and more numerous. And nobody is watching them, because they sit above 4,500 metres and are days of walking from the nearest road. You cannot put sensors on hundreds of lakes in that terrain and keep them working.

2. Why this matters — what happened in Sikkim

On the night of 4 October 2023, South Lhonak Lake in Sikkim burst.

55	50M	1,200	7
people confirmed dead, and 74 more never found	tonnes of water released, in a few hours	megawatt Teesta-III dam completely destroyed	years the warning sign was visible beforehand

> **This is the part that matters for our project.** The hillside that collapsed into the lake had been slowly sliding for seven years — moving more than 15 metres every year. That movement was visible in free satellite photographs the entire time. Scientists had even published a paper in 2021 warning that this exact lake could burst in this exact way.
>
> So the information existed. The warning sign was there, in public data, for years. What was missing was simply **a system that was looking at it**.

That is the gap we are filling. We are not trying to predict the unpredictable. We are trying to watch something that is already visible, and that nobody is currently watching.

3. Problem statement

> **Build a software system that automatically watches 15 to 25 Himalayan glacial lakes using free satellite data, learns what "normal" looks like for each lake, and produces a ranked list of which lakes are behaving abnormally this month — along with the evidence, so that a disaster-management officer can decide what to do.**

Broken into simple parts

- Hundreds of dangerous lakes exist and nobody monitors them regularly.
- Warning signs are visible in free satellite data weeks to months before a lake bursts.

- Nobody has built a system that watches those signs continuously across many lakes.
- During the monsoon — the most dangerous season — clouds block ordinary satellite cameras, so any camera-only system goes blind exactly when it is needed most.

4. The research gap — what has been done, and what has not

What researchers have already done	What it does not do	So the gap is…
Counted and mapped the lakes **Surveys listing how many glacial lakes exist and how fast they are growing.**	It is a headcount taken once. It tells you a lake exists; it does not tell you how that lake is behaving today.	Nobody has built a system that **watches lakes continuously and raises a flag before** anything happens. Everything that exists either counts lakes, or explains a disaster afterwards.
Explained disasters afterwards **Detailed studies reconstructing exactly what happened in Sikkim in 2023.**	Excellent science, but it arrives after the flood. It cannot save anyone.	
Built flood-damage models **Simulations of how far the water would travel and what it would destroy.**	These answer "what if it bursts", not "is it about to burst".	
Started combining satellite data with AI **Two recent works — GLOFNet (2025) and IceWatch (2026) — combine three types of satellite data.**	Both only use ordinary cameras, so both go blind in monsoon cloud. Both cover only the Karakoram, not the Indian Himalaya. And neither can say "I cannot see this lake."	

> **The best part: we did not decide this gap ourselves.** The team that built the GLOFNet dataset wrote, in their own paper, that the next steps should be to add radar, improve the temperature data, and expand to more mountain regions. Those are exactly our objectives. We are building the thing that the previous researchers said should be built next.

5. What we are actually building — the end product

At the end of 14 weeks, there will be a working web dashboard that a disaster-management officer can open in a browser. It has four things on it:

Screen	What the officer sees
1. The watch-list map	A map of all monitored lakes, coloured by risk, with a ranked list beside it — most concerning lake at the top. The rule we set ourselves: an officer should understand this screen in five seconds.
2. The lake detail page	Click any lake and see every signal plotted over time — how the lake's size changed, how fast the glacier moved, how warm it got. Periods where we had no data are shaded grey, so nothing is hidden.
3. Alert settings	The officer decides how sensitive the warning should be for each lake. We do not decide that for them.
4. Evidence report	One click produces a printable PDF: before-and-after satellite pictures, the charts, and the current score — something the officer can carry into a meeting.

> **One rule we will not break.** The system must never say a lake is safe when it simply cannot see it. If clouds block the view, it says **"cannot assess"** — not "low risk". On the screen, a lake we cannot see is never the same colour as a lake that looks fine. "I don't know" and "it's fine" are completely different statements, and mixing them up is the most dangerous mistake this software could make.

Alongside the software we will also produce a research paper, a full project report, and a public code repository that the next batch of students can pick up and continue.

6. How it works — in plain steps

Step	What happens	Explained simply
1	Collect satellite data	Download pictures and measurements of each lake from free public archives, going back to 2017.
2	Line everything up	Different satellites use different scales and dates. We convert them all to one common map grid and one common 10-day calendar, so they can be compared honestly.
3	Turn pictures into numbers	From the imagery we measure real quantities: how big is the lake, how fast is the glacier moving, how warm has it been, has the hillside changed.
4	Learn what normal looks like	For each lake, the computer studies several years of its own history and works out its usual behaviour — including the fact that summers are always warmer.
5	Measure how far from normal it is today	Each lake gets a score. A high score means several signals are drifting away from that lake's own normal at the same time.
6	Show it to a human	Ranked watch-list, with the evidence attached. A person makes the decision. The software never raises a public alarm by itself.

The key idea, and why it is unusual

The obvious approach would be to show a computer many examples of lakes that burst and many that did not, and let it learn the difference. That is how most AI works.

We cannot do that. In India there is essentially one well-documented case — Sikkim 2023. You cannot teach a computer to recognise a pattern from a single example; it would simply memorise that one lake and be useless on any other.

So we flip the problem. Instead of learning what a *bursting* lake looks like, we teach the computer what each lake looks like when it is *normal* — and there are thousands of lake-years of normal data available for free. Then we simply measure how far today is from that normal. The one real disaster is never used for teaching; it is saved as the final exam.

7. The four signals we watch

No single measurement is trustworthy on its own — satellites are noisy. Confidence comes from several independent signals moving together at the same time.

Signal	Satellite	What it tells us, in plain words
Lake growth	Sentinel-2 (a normal colour camera)	Is the lake getting bigger, or is its edge changing shape? A growing lake puts more pressure on the wall holding it back.
Hillside movement	Sentinel-1 (radar — see below)	Is the rock wall around the lake shifting? This is the signal that was visible for seven years before Sikkim.
Glacier speeding up	NASA ITS_LIVE	Glaciers always creep downhill. If one suddenly speeds up, something has changed — and this is what happened before Sikkim.

Signal	Satellite	What it tells us, in plain words
Heat and melting	MODIS	How warm has it been? Warmer weather means more meltwater flowing into the lake.

What is radar, and why does it matter so much here? An ordinary satellite camera is like your eye — if there is cloud in the way, it sees nothing. Radar is different: it sends out its own radio signal and measures what bounces back, so it works through cloud, and at night.

This matters because the monsoon is both the cloudiest season and the most dangerous season. A camera-only system is blind for months at exactly the wrong time. **Adding radar is the main technical contribution of our project**, and measuring exactly how much extra visibility it buys us is one of our two headline results.

8. The data we use — and what it costs

Every dataset in this project is public and free. Governments and space agencies fund these satellites and give the data away. **Our total data bill is zero rupees.**

Dataset	What it gives us	Who provides it	Cost
Sentinel-2	Colour photographs of each lake, detailed enough to see a 10-metre object, every 5 days	European Space Agency	Free
Sentinel-1	Radar images that work through cloud and at night	European Space Agency	Free
ITS_LIVE	How fast every glacier in the world is moving, from the 1980s to now	NASA	Free
MODIS	Daily surface temperature readings	NASA	Free
Copernicus DEM	A 3-D height map of the terrain — how steep the slopes around each lake are	European Space Agency	Free
GLOFNet	A ready-made research dataset combining three of the above, so we can start work in week one	Published research, 2025	Free
HMA GLOF Database	A record of 697 real lake-burst floods between 1833 and 2022 — this is our answer key for testing	Published research, 2023	Free
ICIMOD inventories	Lists of Himalayan glacial lakes, including which ones experts consider dangerous — we use this to choose our study lakes	ICIMOD, Nepal	Free

The only money the project needs is roughly ₹10,000, almost all of it for a backup hard disk, printing and stationery. Not for data, and not for computing.

9. Technology we will use

What for	Tool	Why this one
Writing the software	Python	The standard language for data science and satellite work; every library we need already exists.
Handling satellite images	Google Earth Engine	Google stores the world's satellite archive and lets us process it *on their computers*, free for research. We download only the small final answers instead of enormous image files.

What for	Tool	Why this one
Running our programs	Google Colab	A free online computer with a graphics card, so no student needs an expensive laptop.
Reading and aligning images	rasterio, GDAL, xarray	Standard tools for opening satellite files and stacking many dates into one tidy block of data.
The learning part	scikit-learn, PyTorch	scikit-learn gives us the simpler methods (Isolation Forest, One-Class SVM) that spot unusual behaviour; PyTorch handles the more advanced model that learns each lake's rhythm over time.
The dashboard	Streamlit, Folium, Plotly	Lets us build a proper web dashboard in plain Python — no separate web-development work needed.
Keeping it reproducible	Git, GitHub, MLflow	Every result can be traced back to the exact code and settings that produced it. If we cannot reproduce a number, we do not report it.

10. Who does what

The work is split by function, not by topic, so all four of us are fully occupied from week one. Importantly, **nobody has to wait for anybody else during the first month** — that was designed in deliberately.

Student	Role	What they are responsible for	What they hand over
A	Data Collection	Fetching all the satellite data, cleaning it, removing clouds, lining everything up onto one grid, and adding the radar. Effectively the supply line for the whole team.	One tidy data file that B and C can open in a single line of code
B	Model Development	Turning the raw data into meaningful measurements, then building the models that learn each lake's normal behaviour and score how far it has drifted.	A risk score for every lake, for every 10-day period
C	Testing & Evaluation	Independently checking whether the system actually works. Writes the test rules *before* seeing any results, and runs the Sikkim examination.	Every number and chart that appears in the report
D	App Development	Building the dashboard the officer uses, plus the literature survey, documentation, project report and research paper.	The working app, the report and the paper

Student C deliberately does not build any part of the model. This means the people who build the system are not the same people who judge whether it works — the same reason an exam is not marked by the candidate.

11. How we will prove it actually works

The most important test is simple to explain. We will feed the system **only the data that existed before 4 October 2023** — nothing from the day of the Sikkim disaster or after — and ask: would it have flagged South Lhonak Lake, and how many weeks in advance?

Test	What it answers
The Sikkim examination	Using only pre-disaster data, how much warning would the system have given? This is our headline result.
The cheating check	An automatic test that fails the run if even one piece of after-the-event data sneaks in. We will deliberately feed it cheated data first, to prove the check actually catches it.
The radar benefit	How many extra days of visibility does radar give us during the monsoon, compared to cameras alone? This number does not currently exist in the research literature.
The unseen-lake test	Train on some lakes, then test on lakes the system has never encountered — proving it learned something general rather than memorising.
The which-signals-matter test	Remove one satellite at a time and see how much worse it gets. This tells a real monitoring agency what they actually need to pay for.
The honesty test	Compare every model against the simplest possible method. If the fancy model does not beat the simple one, we report that plainly.

12. What we are deliberately not doing

Being clear about limits is part of doing this responsibly.

We are not building…	Because
A siren that warns the public automatically	This is a tool that helps an expert decide. A false public alarm in a mountain valley causes real harm, so a human stays in the loop at all times.
A minute-by-minute flood warning	We work on a scale of weeks to months — spotting a lake becoming dangerous, not detecting a flood already in progress.
A simulation of the flood itself	Modelling how far the water travels is a different specialisation. We stop at "this lake is behaving abnormally".
A system for every kind of lake	We cover lakes held back by rock-and-mud walls. Lakes dammed by ice fail differently and need different physics.
Anything requiring paid data	A system a state agency cannot afford to run is not a solution. Free data is a design requirement, not a compromise.

13. The plan at a glance

Weeks	What happens	How we will know it worked
1–5	Get the data flowing. Build the simplest possible detector and a skeleton dashboard.	Data for all lakes in one place; a basic score being produced
6	**Checkpoint 1 — go or no-go.** The whole chain must work end to end on real data.	Real scores exist and appear on the dashboard
7–10	Proper models. Add radar. Run the Sikkim examination.	The two headline results are produced
8	**Checkpoint 2 — mid-project demonstration.**	Dashboard runs live in front of the guide
11–13	Which-signals-matter test, unseen-lake test, report and paper.	All results locked; nothing changes after week 13

Weeks	What happens	How we will know it worked
14	Deliberately kept empty as a safety margin.	Paper submitted, demonstration rehearsed

14. Questions the panel may ask — short answers

How can you train a model when only one disaster has ever been properly recorded?

We never train on the disaster at all. We train on normal behaviour, which is abundant and free, and measure how far a lake has drifted from its own normal. The Sikkim event is kept aside as the final exam.

Why not just install sensors on the lakes?

They are above 4,500 metres and days from a road. You cannot install and maintain equipment on hundreds of them. Satellites already visit every lake every few days, for free.

What if your system does not find anything?

The project still delivers. We would produce the first multi-satellite dataset for Indian Himalayan lakes, a measurement of what radar adds during the monsoon, and an honest account of why this is hard. A negative result that is properly measured is a real scientific contribution — and we would rather report that than claim something we cannot support.

How is this different from the recent papers doing the same thing?

Three ways. They use only cameras, so they are blind in monsoon cloud — we add radar. They work in the Karakoram — we work in the Indian Himalaya. And they always produce a score, whereas ours can say "I cannot see this lake", which for a safety system is essential.

What will it cost to run?

Nothing. Every dataset is public and every computing platform we use is a free tier. A state agency could adopt it without a budget approval.

Who would actually use this?

A State Disaster Management Authority early-warning officer, in Sikkim or Uttarakhand. Also hydropower operators on Himalayan rivers — the Teesta-III station destroyed in 2023 is exactly the kind of asset at risk.

> **The line to close on:** the warning signs were visible in free satellite data for seven years before Sikkim. The information existed. No system was watching it. We are building the system that watches.

Sources for the figures on page 1: Sattar et al., *Science* (2025), doi 10.1126/science.ads2659 · Sattar et al., *Geomorphology* 388 (2021) — the paper that modelled this exact failure two years before it happened · Zhang, Wang & An, *Landslides* 22 (2025). Companion documents: Project Synopsis, SRS, Software Design Document, Resource Pack and four Student Handbooks.